2.14.5 Summing the BCH Series

Inserting these terms into the BCH expansion:

$$\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \hat{a} - (\xi\hat{a}^\dagger) + \frac{1}{2!}(r^2\hat{a}) - \frac{1}{3!}(r^2\xi\hat{a}^\dagger) + \frac{1}{4!}(r^4\hat{a}) - \dots$$

Grouping by $\hat{a}$ and $\hat{a}^\dagger$:

$$\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \hat{a}\left(1 + \frac{r^2}{2!} + \frac{r^4}{4!} + \dots\right) - \xi\hat{a}^\dagger\left(1 + \frac{r^2}{3!} + \frac{r^4}{5!} + \dots\right)$$

With $\xi = re^{i\theta}$, factoring out r from the second series gives:

$$\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \hat{a}\left(1 + \frac{r^2}{2!} + \frac{r^4}{4!} + \dots\right) - e^{i\theta}\hat{a}^\dagger\left(r + \frac{r^3}{3!} + \frac{r^5}{5!} + \dots\right)$$

This matches the Taylor series for $\cosh(r)$ and $\sinh(r)$:

$$\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \cosh(r)\hat{a} - e^{i\theta}\sinh(r)\hat{a}^\dagger$$

2.14.6 Final Expression

Defining $\mu = \cosh r$ and $\nu = e^{i\theta}\sinh r$:

$$\hat{S}(\xi)^\dagger\hat{a}\hat{S}(\xi) = \mu\hat{a} - \nu\hat{a}^\dagger \tag{2.71}$$

2.15 Derivation of the Normal-Ordered Squeezing Operator using Operator Algebra

Our goal is to rewrite the single-mode squeezing operator $\hat{S}(\xi)$ into normal order, where all creation operators ($\hat{a}^\dagger$) are on the left, the number operators ($\hat{n} = \hat{a}^\dagger\hat{a}$) are in the middle, and the annihilation operators ($\hat{a}$) are on the right.

2.15.1 Constructing the Normal-Ordered Ansatz

Let us propose a general normal-ordered ansatz with unknown scalar coefficients A, B, and C, and a normalization constant M:

$$\hat{S} = Me^{A(\hat{a}^\dagger)^2}e^{B\hat{a}^\dagger\hat{a}}e^{C\hat{a}^2}$$

Because the squeezing operator is unitary, its inverse is its adjoint ($\hat{S}^{-1} = \hat{S}^\dagger$). From

the work above, we know how this operator transforms the fundamental mode operators:

$$\hat{S}^{-1}\hat{a}\hat{S} = \mu\hat{a} - \nu\hat{a}^\dagger$$
$$\hat{S}^{-1}\hat{a}^\dagger\hat{S} = \mu\hat{a}^\dagger - \nu^*\hat{a}$$

where $\mu = \cosh r$ and $\nu = e^{i\theta}\sinh r$.

If we push the operators $\hat{a}$ and $\hat{a}^\dagger$ through our ansatz using the Baker-Campbell-Hausdorff (BCH) lemma and equate the results to the known transformations, we can solve for A, B, and C.

2.15.2 Pushing the Creation Operator Through the Ansatz

We evaluate the transformation of $\hat{a}^\dagger$ using our proposed form:

$$\hat{S}^{-1}\hat{a}^\dagger\hat{S} = \left(e^{-C\hat{a}^2}e^{-B\hat{a}^\dagger\hat{a}}e^{-A(\hat{a}^\dagger)^2}M^{-1}\right)\hat{a}^\dagger\left(Me^{A(\hat{a}^\dagger)^2}e^{B\hat{a}^\dagger\hat{a}}e^{C\hat{a}^2}\right)$$

The scalars M and M^{-1} cancel out. We move $\hat{a}^\dagger$ from the center to the right using the BCH expansion: $e^{\hat{X}}\hat{Y}e^{-\hat{X}} = \hat{Y} + [\hat{X},\hat{Y}] + \frac{1}{2!}[\hat{X},[\hat{X},\hat{Y}]] + \dots$

Step 2a: Commute through the A term
We evaluate $e^{-A(\hat{a}^\dagger)^2}\hat{a}^\dagger e^{A(\hat{a}^\dagger)^2}$. Because $[\hat{a}^\dagger,\hat{a}^\dagger] = 0$, all commutators are zero. The operator passes through unchanged:

$$\hat{a}^\dagger$$

Step 2b: Commute through the B term
We evaluate $e^{-B\hat{n}}\hat{a}^\dagger e^{B\hat{n}}$. Using $\hat{X} = -B\hat{n}$ and $\hat{Y} = \hat{a}^\dagger$, and knowing $[\hat{n},\hat{a}^\dagger] = \hat{a}^\dagger$:

$$[-B\hat{n},\hat{a}^\dagger] = -B\hat{a}^\dagger$$
$$[-B\hat{n},-B\hat{a}^\dagger] = B^2\hat{a}^\dagger$$

Summing the series gives:

$$\hat{a}^\dagger\left(1 - B + \frac{B^2}{2!} - \dots\right) = e^{-B}\hat{a}^\dagger$$

Step 2c: Commute through the C term
We evaluate $e^{-C\hat{a}^2}(e^{-B}\hat{a}^\dagger)e^{C\hat{a}^2}$. Using $\hat{X} = -C\hat{a}^2$ and $\hat{Y} = \hat{a}^\dagger$, and knowing $[\hat{a}^2,\hat{a}^\dagger] = 2\hat{a}$:

$$[-C\hat{a}^2,\hat{a}^\dagger] = -C(2\hat{a}) = -2C\hat{a}$$
$$[-C\hat{a}^2,-2C\hat{a}] = 2C^2[\hat{a}^2,\hat{a}] = 0$$

The series truncates immediately, yielding:

$$e^{-B}(\hat{a}^\dagger - 2C\hat{a})$$

Solving for B and C:
We equate our final expression to the known transformation:

$$e^{-B}\hat{a}^\dagger - 2Ce^{-B}\hat{a} = \mu\hat{a}^\dagger - \nu^*\hat{a}$$

Matching the coefficients of $\hat{a}^\dagger$:

$$e^{-B} = \mu \implies B = -\ln\mu$$

Matching the coefficients of $\hat{a}$:

$$-2C(e^{-B}) = -\nu^* \implies -2C\mu = -\nu^* \implies C = \frac{\nu^*}{2\mu}$$

2.15.3 Pushing the Annihilation Operator Through the Ansatz

Now we evaluate the transformation of $\hat{a}$ to find A:

$$\hat{S}^{-1}\hat{a}\hat{S} = e^{-C\hat{a}^2}e^{-B\hat{a}^\dagger\hat{a}}e^{-A(\hat{a}^\dagger)^2}\hat{a}e^{A(\hat{a}^\dagger)^2}e^{B\hat{a}^\dagger\hat{a}}e^{C\hat{a}^2}$$

Step 3a: Commute through the A term
Evaluate $e^{-A(\hat{a}^\dagger)^2}\hat{a}e^{A(\hat{a}^\dagger)^2}$. Knowing $[(\hat{a}^\dagger)^2, \hat{a}] = -2\hat{a}^\dagger$:

$$\hat{a} + 2A\hat{a}^\dagger$$

Step 3b: Commute through the B term
Evaluate $e^{-B\hat{n}}(\hat{a} + 2A\hat{a}^\dagger)e^{B\hat{n}}$. We apply the linear transformation to both terms. From Step 2b, $\hat{a}^\dagger \to e^{-B}\hat{a}^\dagger$. For the $\hat{a}$ term, knowing $[\hat{n}, \hat{a}] = -\hat{a}$, the BCH series evaluates to $e^B\hat{a}$. The result is:

$$e^B\hat{a} + 2Ae^{-B}\hat{a}^\dagger$$

Step 3c: Commute through the C term
Evaluate $e^{-C\hat{a}^2}(e^B\hat{a} + 2Ae^{-B}\hat{a}^\dagger)e^{C\hat{a}^2}$. Since $[\hat{a}^2, \hat{a}] = 0$, the first term is untouched. From Step 2c, $\hat{a}^\dagger \to \hat{a}^\dagger - 2C\hat{a}$.

$$e^B\hat{a} + 2Ae^{-B}(\hat{a}^\dagger - 2C\hat{a}) = (e^B - 4ACe^{-B})\hat{a} + 2Ae^{-B}\hat{a}^\dagger$$

Solving for A:

Equate the coefficient of $\hat{a}^\dagger$ to the known transformation $(\mu\hat{a} - \nu\hat{a}^\dagger)$:

$$2Ae^{-B} = -\nu \implies 2A\mu = -\nu \implies A = -\frac{\nu}{2\mu}$$

2.15.4 Determining the Normalization Scalar

Our ansatz is now populated with the operational coefficients:

$$\hat{S} = M \exp\left(-\frac{\nu}{2\mu}(\hat{a}^\dagger)^2\right) \exp\left(-(\ln\mu)\hat{a}^\dagger\hat{a}\right) \exp\left(\frac{\nu^*}{2\mu}\hat{a}^2\right)$$

We apply this operator to the vacuum state $|0\rangle$ to enforce the norm condition $\langle\xi|\xi\rangle = 1$, where $|\xi\rangle = \hat{S}|0\rangle$. Because $\hat{a}^2|0\rangle = 0$ and $\hat{a}^\dagger\hat{a}|0\rangle = 0$, the two rightmost exponentials act as the identity:

$$|\xi\rangle = M \exp\left(-\frac{\nu}{2\mu}(\hat{a}^\dagger)^2\right)|0\rangle$$

Expanding this into a Taylor series and applying it to the vacuum yields the Fock state expansion:

$$|\xi\rangle = M \sum_{k=0}^{\infty} \frac{1}{k!}\left(-\frac{\nu}{2\mu}\right)^k \sqrt{(2k)!}|2k\rangle$$

Squaring this to find the total probability (the norm), which must equal 1:

$$\langle\xi|\xi\rangle = |M|^2 \sum_{k=0}^{\infty} \frac{(2k)!}{(k!)^2}\left|\frac{\nu}{2\mu}\right|^{2k} = 1$$

Using the binomial series for negative fractional powers, $(1-x)^{-1/2} = \sum_{k=0}^{\infty} \frac{(2k)!}{(k!)^2}\left(\frac{x}{4}\right)^k$, we substitute $x = \tanh^2 r$ (since $\frac{\tanh^2 r}{4} = \left|\frac{\nu}{2\mu}\right|^2$).

$$\sum_{k=0}^{\infty} \frac{(2k)!}{(k!)^2}\left|\frac{\nu}{2\mu}\right|^{2k} = (1-\tanh^2 r)^{-1/2} = (\mathrm{sech}^2 r)^{-1/2} = \cosh r = \mu$$

Substituting this back into the norm equation:

$$|M|^2\mu = 1 \implies M = \mu^{-1/2}$$

2.15.5 Final Assembly

We place M back into the full operational expression:

$$\hat{S}(\xi) = \mu^{-1/2} \exp\left(-\frac{\nu}{2\mu}(\hat{a}^\dagger)^2\right) \exp\left(-(\ln\mu)\hat{a}^\dagger\hat{a}\right) \exp\left(\frac{\nu^*}{2\mu}\hat{a}^2\right)$$

We combine the scalar $\mu^{-1/2}$ into the middle exponent using logarithm rules ($\mu^y = e^{y \ln \mu}$):

$$\mu^{-1/2} \exp\left(-(\ln \mu)\hat{a}^\dagger \hat{a}\right) = \mu^{-1/2}\mu^{-\hat{a}^\dagger \hat{a}} = \mu^{-(\hat{a}^\dagger \hat{a}+1/2)}$$

Yielding the final, exact normal-ordered form of the squeezing operator:

$$\hat{S}(\xi) = \exp\left(-\frac{\nu}{2\mu}(\hat{a}^\dagger)^2\right)\mu^{-(\hat{a}^\dagger \hat{a}+\frac{1}{2})}\exp\left(\frac{\nu^*}{2\mu}\hat{a}^2\right) \tag{2.72}$$

2.15.6 Squeezed state form

We have stated earlier:

$$|\xi\rangle = M\sum_{k=0}^{\infty}\frac{1}{k!}\left(-\frac{\nu}{2\mu}\right)^k\sqrt{(2k)!}|2k\rangle$$

Thus, we have

$$|\xi\rangle = \mu^{-1/2}\sum_{k=0}^{\infty}\frac{1}{k!}\left(-\frac{\nu}{2\mu}\right)^k\sqrt{(2k)!}|2k\rangle \tag{2.73}$$

which is also a consequent result of our ansatz based approach.

2.16 Derivation of the Wigner Function for the Squeezed Vacuum State

This derivation approaches the Wigner function $W(x,p)$ of the squeezed vacuum state by first determining its position wavefunction, and then applying the Fourier transform definition of the Wigner function.

2.16.1 The Annihilation Condition

For a state created by acting on the vacuum $|0\rangle$ with a unitary operator, the resulting state is the "vacuum" for the correspondingly transformed annihilation operator.

For the squeezed vacuum state $|\xi\rangle = \hat{S}(\xi)|0\rangle$, we define a transformed annihilation operator $\hat{b}$:

$$\hat{b} = \hat{S}(\xi)\hat{a}\hat{S}^\dagger(\xi)$$

Because the standard annihilation operator destroys the vacuum ($\hat{a}|0\rangle = 0$), the transformed operator must destroy the squeezed vacuum:

$$\hat{b}|\xi\rangle = \hat{S}(\xi)\hat{a}\hat{S}^\dagger(\xi)\hat{S}(\xi)|0\rangle = \hat{S}(\xi)\hat{a}|0\rangle = 0$$

2.16.2 Formulating $\hat{b}$ in Phase Space

From the known mode evolution, $\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \mu\hat{a} - \nu\hat{a}^\dagger$. To find $\hat{S}(\xi)\hat{a}\hat{S}^\dagger(\xi)$, we invert the operation by replacing the squeezing parameter ξ with $-\xi$, which flips the sign of ν:

$$\hat{b} = \mu\hat{a} + \nu\hat{a}^\dagger$$

Assuming a purely real squeezing parameter $\xi = r$ (setting the angle $\theta = 0$), we have $\mu = \cosh r$ and $\nu = \sinh r$:

$$\hat{b} = \cosh(r)\hat{a} + \sinh(r)\hat{a}^\dagger$$

We express the creation and annihilation operators in terms of the position and momentum quadratures, $\hat{a} = \hat{q} + i\hat{p}$ and $\hat{a}^\dagger = \hat{q} - i\hat{p}$:

$$\begin{aligned}\hat{b} &= \cosh(r)(\hat{q} + i\hat{p}) + \sinh(r)(\hat{q} - i\hat{p}) \\ &= (\cosh r + \sinh r)\hat{q} + i(\cosh r - \sinh r)\hat{p}\end{aligned}$$

Using the identities $e^{\pm r} = \cosh r \pm \sinh r$, we finalize the operator:

$$\hat{b} = e^r\hat{q} + ie^{-r}\hat{p}$$

2.16.3 Solving the Position Wavefunction

We apply the condition $\hat{b}|\xi\rangle = 0$ in the continuous position basis to solve for the wavefunction $\psi(x) = \langle x|\xi\rangle$.

The commutation relation $[\hat{q}, \hat{p}] = i/2$ dictates the representations $\hat{q} \to x$ and $\hat{p} \to -\frac{i}{2}\frac{d}{dx}$. Substituting these yields a first-order differential equation:

$$\begin{aligned}\left(e^r x + ie^{-r}\left(-\frac{i}{2}\frac{d}{dx}\right)\right)\psi(x) &= 0 \\ e^r x\psi(x) + \frac{1}{2}e^{-r}\frac{d\psi(x)}{dx} &= 0 \\ \frac{d\psi(x)}{dx} &= -2e^{2r}x\psi(x)\end{aligned}$$

Integrating this separable differential equation produces a Gaussian distribution with a normalization constant N:

$$\begin{aligned}\ln\psi(x) &= -e^{2r}x^2 + C \\ \psi(x) &= N\exp\left(-e^{2r}x^2\right)\end{aligned}$$

We find N by enforcing the probability normalization $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$:

$$|N|^2 \int_{-\infty}^{\infty} \exp\left(-2e^{2r}x^2\right) dx = 1$$

Using the standard Gaussian integral $\int e^{-ax^2} dx = \sqrt{\pi/a}$ with $a = 2e^{2r}$:

$$|N|^2 \sqrt{\frac{\pi}{2e^{2r}}} = 1 \implies N = \left(\frac{2e^{2r}}{\pi}\right)^{1/4}$$

The exact, normalized position wavefunction is:

$$\psi(x) = \left(\frac{2e^{2r}}{\pi}\right)^{1/4} \exp\left(-e^{2r}x^2\right)$$

2.16.4 Integrating the Wigner Function

The Wigner function for a pure state is:

$$W(x,p) = \frac{1}{\pi\hbar} \int_{-\infty}^{\infty} \psi^*(x+y)\psi(x-y)e^{2ipy/\hbar} dy$$

Because $[\hat{q}, \hat{p}] = i/2$, the effective value for $\hbar$ is $1/2$. This simplifies the expression to:

$$W(x,p) = \frac{2}{\pi} \int_{-\infty}^{\infty} \psi^*(x+y)\psi(x-y)e^{4ipy} dy$$

We explicitly evaluate the product of the wavefunctions:

$$\begin{aligned}\psi^*(x+y)\psi(x-y) &= \sqrt{\frac{2e^{2r}}{\pi}} \exp\left[-e^{2r}(x+y)^2 - e^{2r}(x-y)^2\right] \\ &= \sqrt{\frac{2e^{2r}}{\pi}} \exp\left(-2e^{2r}x^2 - 2e^{2r}y^2\right)\end{aligned}$$

Substituting this back into the Wigner integral and pulling independent terms outside:

$$W(x,p) = \frac{2}{\pi}\sqrt{\frac{2e^{2r}}{\pi}} \exp\left(-2e^{2r}x^2\right) \int_{-\infty}^{\infty} \exp\left(-2e^{2r}y^2\right) e^{4ipy} dy$$

The integral is a standard Fourier transform of a Gaussian, $\int e^{-ay^2+by} dy = \sqrt{\frac{\pi}{a}} e^{b^2/4a}$, with $a = 2e^{2r}$ and $b = 4ip$:

$$\frac{b^2}{4a} = \frac{(4ip)^2}{4(2e^{2r})} = \frac{-16p^2}{8e^{2r}} = -2e^{-2r}p^2$$

The integral evaluates to:

$$\sqrt{\frac{\pi}{2e^{2r}}}\exp\left(-2e^{-2r}p^2\right)$$

Multiplying this by the prefactors:

$$W(x,p) = \frac{2}{\pi}\sqrt{\frac{2e^{2r}}{\pi}}\exp\left(-2e^{2r}x^2\right)\sqrt{\frac{\pi}{2e^{2r}}}\exp\left(-2e^{-2r}p^2\right)$$

The square root multipliers cancel out exactly ($\sqrt{A/\pi}\cdot\sqrt{\pi/A}=1$). The final Wigner function is:

$$W(x,p) = \frac{2}{\pi}\exp\left(-2e^{2r}x^2 - 2e^{-2r}p^2\right) \tag{2.74}$$

2.17 Algebraic Derivation of Two-Mode Squeezing

2.17.1 Definitions

We would now like to move from the single-mode formalism to two independent bosonic modes, which is required for the generation of continuous-variable entanglement. Let $\hat{a}$ and $\hat{b}$ represent independent modes.

$$[\hat{a},\hat{a}^\dagger] = 1, \quad [\hat{b},\hat{b}^\dagger] = 1$$
$$[\hat{a},\hat{b}] = [\hat{a},\hat{b}^\dagger] = [\hat{a}^\dagger,\hat{b}] = 0$$

The two-mode squeezing operator is defined as:

$$\hat{S}_2(\xi) = \exp\Big(\xi\hat{a}^\dagger\hat{b}^\dagger - \xi^*\hat{a}\hat{b}\Big)$$

where $\xi = re^{i\theta}$.

2.17.2 Mode Evolution

We evaluate $\hat{S}_2^\dagger(\xi)\hat{a}\hat{S}_2(\xi)$. Define the exponent $\hat{K} = \xi^*\hat{a}\hat{b} - \xi\hat{a}^\dagger\hat{b}^\dagger$. Using the Baker-Campbell-Hausdorff (BCH) (similar to the one for single mode):

$$e^{\hat{K}}\hat{a}e^{-\hat{K}} = \hat{a} + [\hat{K},\hat{a}] + \frac{1}{2!}[\hat{K},[\hat{K},\hat{a}]] + \dots$$

Evaluating the commutators:

$$[\hat{K},\hat{a}] = [\xi^*\hat{a}\hat{b} - \xi\hat{a}^\dagger\hat{b}^\dagger,\hat{a}] = [-\xi\hat{a}^\dagger\hat{b}^\dagger,\hat{a}] = -\xi[\hat{a}^\dagger,\hat{a}]\hat{b}^\dagger = \xi\hat{b}^\dagger$$
$$[\hat{K},[\hat{K},\hat{a}]] = [\xi^*\hat{a}\hat{b} - \xi\hat{a}^\dagger\hat{b}^\dagger,\xi\hat{b}^\dagger] = [\xi^*\hat{a}\hat{b},\xi\hat{b}^\dagger] = \xi\xi^*\hat{a}[\hat{b},\hat{b}^\dagger] = r^2\hat{a}$$

Substituting back into the series:

$$\hat{S}_2^\dagger \hat{a} \hat{S}_2 = \hat{a}\left(1 + \frac{r^2}{2!} + \frac{r^4}{4!} + \dots\right) + \xi \hat{b}^\dagger \left(1 + \frac{r^2}{3!} + \frac{r^4}{5!} + \dots\right)$$

Substituting $\xi = e^{i\theta} r$, $\mu = \cosh r$, and $\nu = e^{i\theta} \sinh r$:

$$\hat{S}_2^\dagger \hat{a} \hat{S}_2 = \mu \hat{a} + \nu \hat{b}^\dagger \tag{2.75}$$

By symmetry for mode $\hat{b}$:

$$\hat{S}_2^\dagger \hat{b}^\dagger \hat{S}_2 = \nu^* \hat{a} + \mu \hat{b}^\dagger \tag{2.76}$$

2.17.3 Normal-Ordered Ladder Expansion

Propose an algebraic ansatz with coefficients A, B, C and scalar M:

$$\hat{S}_2 = M e^{A\hat{a}^\dagger \hat{b}^\dagger} e^{B(\hat{a}^\dagger \hat{a} + \hat{b}^\dagger \hat{b})} e^{C\hat{a}\hat{b}}$$

The inverse is:

$$\hat{S}_2^{-1} = e^{-C\hat{a}\hat{b}} e^{-B(\hat{n}_a + \hat{n}_b)} e^{-A\hat{a}^\dagger \hat{b}^\dagger} M^{-1}$$

Step 3a: Push $\hat{b}^\dagger$ through $\hat{S}_2^{-1}$

$$\begin{aligned}
&\text{Through A:} \quad e^{-A\hat{a}^\dagger \hat{b}^\dagger} \hat{b}^\dagger e^{A\hat{a}^\dagger \hat{b}^\dagger} = \hat{b}^\dagger \\
&\text{Through B:} \quad e^{-B\hat{n}_b} \hat{b}^\dagger e^{B\hat{n}_b} \implies [-B\hat{n}_b, \hat{b}^\dagger] = -B\hat{b}^\dagger \implies e^{-B}\hat{b}^\dagger \\
&\text{Through C:} \quad e^{-C\hat{a}\hat{b}}(e^{-B}\hat{b}^\dagger)e^{C\hat{a}\hat{b}} \implies [-C\hat{a}\hat{b}, \hat{b}^\dagger] = -C\hat{a} \implies e^{-B}(\hat{b}^\dagger - C\hat{a})
\end{aligned}$$

Equating to the known evolution $\hat{S}_2^{-1}\hat{b}^\dagger \hat{S}_2 = \nu^* \hat{a} + \mu \hat{b}^\dagger$:

$$e^{-B}\hat{b}^\dagger - Ce^{-B}\hat{a} = \mu \hat{b}^\dagger + \nu^* \hat{a} \implies B = -\ln \mu, \quad C = -\frac{\nu^*}{\mu}$$

Step 3b: Push $\hat{a}$ through $\hat{S}_2^{-1}$

$$\begin{aligned}
&\text{Through A:} \quad e^{-A\hat{a}^\dagger \hat{b}^\dagger} \hat{a} e^{A\hat{a}^\dagger \hat{b}^\dagger} \implies [-A\hat{a}^\dagger \hat{b}^\dagger, \hat{a}] = A\hat{b}^\dagger \implies \hat{a} + A\hat{b}^\dagger \\
&\text{Through B:} \quad e^{-B(\hat{n}_a + \hat{n}_b)}(\hat{a} + A\hat{b}^\dagger)e^{B(\hat{n}_a + \hat{n}_b)} \implies e^B \hat{a} + Ae^{-B}\hat{b}^\dagger \\
&\text{Through C:} \quad e^{-C\hat{a}\hat{b}}(e^B \hat{a} + Ae^{-B}\hat{b}^\dagger)e^{C\hat{a}\hat{b}} \implies e^B \hat{a} + Ae^{-B}(\hat{b}^\dagger - C\hat{a}) \\
&\qquad\qquad = (e^B - ACe^{-B})\hat{a} + Ae^{-B}\hat{b}^\dagger
\end{aligned}$$

Equating to $\mu \hat{a} + \nu \hat{b}^\dagger$:

$$Ae^{-B} = \nu \implies A\mu = \nu \implies A = \frac{\nu}{\mu}$$

Step 3c: Determining Normalization M Applying the populated ansatz to the vacuum $|0,0\rangle$, we have:

$$|EPR\rangle = \hat{S}_2|0,0\rangle = M \exp\left(\frac{\nu}{\mu}\hat{a}^\dagger\hat{b}^\dagger\right)|0,0\rangle = M\sum_{k=0}^{\infty}\left(\frac{\nu}{\mu}\right)^k |k,k\rangle$$

Enforce normalization $\langle\psi|\psi\rangle = 1$:

$$\begin{aligned}|M|^2\sum_{k=0}^{\infty}\left|\frac{\nu}{\mu}\right|^{2k} &= |M|^2\sum_{k=0}^{\infty}(\tanh^2 r)^k \\ &= |M|^2\left(\frac{1}{1-\tanh^2 r}\right) = |M|^2\cosh^2 r = 1 \implies M = \mu^{-1}\end{aligned}$$

Step 3d: Final Assembly Substitute M and combine the middle exponent terms using $\mu^y = e^{y\ln\mu}$:

$$\hat{S}_2(\xi) = \mu^{-1}\exp\left(\frac{\nu}{\mu}\hat{a}^\dagger\hat{b}^\dagger\right)\exp\left(-(\ln\mu)(\hat{a}^\dagger\hat{a}+\hat{b}^\dagger\hat{b})\right)\exp\left(-\frac{\nu^*}{\mu}\hat{a}\hat{b}\right) \tag{2.77}$$

$$\hat{S}_2(\xi) = \exp\left(\frac{\nu}{\mu}\hat{a}^\dagger\hat{b}^\dagger\right)\mu^{-(\hat{a}^\dagger\hat{a}+\hat{b}^\dagger\hat{b}+1)}\exp\left(-\frac{\nu^*}{\mu}\hat{a}\hat{b}\right) \tag{2.78}$$

2.17.4 Sidenote :

From the normalization step for M, we can directly substitute and arrive at the state eqn. :

$$|EPR\rangle = \mu^{-1}\sum_{k=0}^{\infty}\left(\frac{\nu}{\mu}\right)^k |k,k\rangle \tag{2.79}$$

2.18 Creating Gaussian EPR states

2.18.1 Preliminary Properties for Usage

Our aim here is to evaluate the expectation values of the mode operators for the squeezed vacuum state $|\xi\rangle = \hat{S}(\xi)|0\rangle$. We shall be relying on the known evolutions: $\hat{S}^\dagger\hat{a}\hat{S} = \mu\hat{a} - \nu\hat{a}^\dagger$ and $\hat{S}^\dagger\hat{a}^\dagger\hat{S} = \mu\hat{a}^\dagger - \nu^*\hat{a}$, where $\mu = \cosh r$ and $\nu = e^{i\theta}\sinh r$.

The core rule for these derivations is that the annihilation operator destroys the vacuum ($\hat{a}|0\rangle = 0$), and its dual creates zero when acting to the left ($\langle 0|\hat{a}^\dagger = 0$).

2.18.2 The First Moments (Means)

The expectation values of the fundamental mode operators evaluate to strictly zero:

$$\begin{aligned}\langle\xi|\hat{a}|\xi\rangle &= \langle 0|\hat{S}^\dagger\hat{a}\hat{S}|0\rangle \\ &= \langle 0|(\mu\hat{a} - \nu\hat{a}^\dagger)|0\rangle \\ &= \mu\langle 0|\hat{a}|0\rangle - \nu\langle 0|\hat{a}^\dagger|0\rangle\end{aligned}$$

Because $\hat{a}|0\rangle = 0$ and $\langle 0|\hat{a}^\dagger = 0$, both terms independently evaluate to zero:

$$\langle\xi|\hat{a}|\xi\rangle = 0$$

Applying the same logic and taking the complex conjugate, the creation operator mean is also zero:

$$\langle\xi|\hat{a}^\dagger|\xi\rangle = 0$$

2.18.3 The Photon Number

To evaluate the expected photon number $\langle\hat{n}\rangle = \langle\hat{a}^\dagger\hat{a}\rangle$, insert the identity $\hat{I} = \hat{S}\hat{S}^\dagger$ between the operators:

$$\begin{aligned}\langle\xi|\hat{a}^\dagger\hat{a}|\xi\rangle &= \langle 0|\hat{S}^\dagger\hat{a}^\dagger(\hat{S}\hat{S}^\dagger)\hat{a}\hat{S}|0\rangle \\ &= \langle 0|(\hat{S}^\dagger\hat{a}^\dagger\hat{S})(\hat{S}^\dagger\hat{a}\hat{S})|0\rangle \\ &= \langle 0|(\mu\hat{a}^\dagger - \nu^*\hat{a})(\mu\hat{a} - \nu\hat{a}^\dagger)|0\rangle\end{aligned}$$

Expanding this binomial product yields:

$$= \langle 0|\mu^2\hat{a}^\dagger\hat{a} - \mu\nu\hat{a}^\dagger\hat{a}^\dagger - \nu^*\mu\hat{a}\hat{a} + |\nu|^2\hat{a}\hat{a}^\dagger|0\rangle$$

Evaluating the action of these four operators on the vacuum state $|0\rangle$:

1. $\hat{a}^\dagger\hat{a}|0\rangle = \hat{a}^\dagger(0) = 0$
2. $\hat{a}^\dagger\hat{a}^\dagger|0\rangle = \sqrt{2}|2\rangle$. Because $\langle 0|2\rangle = 0$, this term vanishes.
3. $\hat{a}\hat{a}|0\rangle = \hat{a}(0) = 0$
4. $\hat{a}\hat{a}^\dagger|0\rangle = \hat{a}|1\rangle = |0\rangle$. This is the only term that survives.

Substituting these evaluations back into the expansion gives:

$$\langle\xi|\hat{a}^\dagger\hat{a}|\xi\rangle = |\nu|^2\langle 0|0\rangle = |\nu|^2$$

Because $\nu = e^{i\theta} \sinh r$, its absolute square is $|\nu|^2 = \sinh^2 r$.

$$\langle \xi | \hat{a}^\dagger \hat{a} | \xi \rangle = \sinh^2 r \tag{2.80}$$

2.18.4 The Variance / Squeezing Terms

The expectations of the paired annihilation and creation operators are calculated similarly:

$$\begin{aligned} \langle \xi | \hat{a}\hat{a} | \xi \rangle &= \langle 0 | (\hat{S}^\dagger \hat{a} \hat{S})(\hat{S}^\dagger \hat{a} \hat{S}) | 0 \rangle \\ &= \langle 0 | (\mu \hat{a} - \nu \hat{a}^\dagger)(\mu \hat{a} - \nu \hat{a}^\dagger) | 0 \rangle \end{aligned}$$

Expanding the binomial product:

$$= \langle 0 | \mu^2 \hat{a}\hat{a} - \mu\nu \hat{a}\hat{a}^\dagger - \nu\mu \hat{a}^\dagger \hat{a} + \nu^2 \hat{a}^\dagger \hat{a}^\dagger | 0 \rangle$$

Applying the vacuum evaluations, only the $\hat{a}\hat{a}^\dagger$ operator survives acting on $|0\rangle$:

$$\langle \xi | \hat{a}\hat{a} | \xi \rangle = -\mu\nu \langle 0 | \hat{a}\hat{a}^\dagger | 0 \rangle = -\mu\nu$$

Substituting the definitions for μ and ν:

$$\langle \xi | \hat{a}\hat{a} | \xi \rangle = -(\cosh r)(e^{i\theta} \sinh r) = -e^{i\theta} \sinh r \cosh r \tag{2.81}$$

Applying identical logic to the creation operators yields $\langle \hat{a}^\dagger \hat{a}^\dagger \rangle$:

$$\begin{aligned} \langle \xi | \hat{a}^\dagger \hat{a}^\dagger | \xi \rangle &= \langle 0 | (\mu \hat{a}^\dagger - \nu^* \hat{a})(\mu \hat{a}^\dagger - \nu^* \hat{a}) | 0 \rangle \\ &= \langle 0 | \mu^2 \hat{a}^\dagger \hat{a}^\dagger - \mu\nu^* \hat{a}^\dagger \hat{a} - \nu^* \mu \hat{a}\hat{a}^\dagger + (\nu^*)^2 \hat{a}\hat{a} | 0 \rangle \\ &= -\mu\nu^* \langle 0 | \hat{a}\hat{a}^\dagger | 0 \rangle = -\mu\nu^* \end{aligned}$$

Substituting the definitions provides the final statistic:

$$\langle \xi | \hat{a}^\dagger \hat{a}^\dagger | \xi \rangle = -e^{-i\theta} \sinh r \cosh r \tag{2.82}$$

2.19 Explicit Derivation of the Matrices for the Gaussian EPR State

The creation of a Gaussian EPR state involves passing two independent squeezed vacuum states through a balanced 50:50 beam-splitter. We construct the three fundamental matrices governing this evolution in phase space: the input covariance matrix (σ_{in}), the symplectic beam-splitter matrix (B), and the output covariance matrix (σ_{EPR}).

2.19.1 Deriving the Input Covariance Matrix

The continuous variable phase space is defined by the quadrature vector $\vec{R} = (\hat{q}_1, \hat{p}_1, \hat{q}_2, \hat{p}_2)^T$. The elements of the covariance matrix σ for zero-mean states are given by $\sigma_{ij} = \frac{1}{2}\langle \hat{R}_i \hat{R}_j + \hat{R}_j \hat{R}_i \rangle$.

We decompose the quadratures into annihilation and creation operators using $\hat{q} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger)$ and $\hat{p} = \frac{1}{2i}(\hat{a} - \hat{a}^\dagger)$. For a single-mode squeezed state (assuming $\theta = 0$), the known expectations are:

$$\langle \hat{a}^\dagger \hat{a} \rangle = \sinh^2 r, \quad \langle \hat{a}\hat{a}^\dagger \rangle = \cosh^2 r, \quad \langle \hat{a}^2 \rangle = \langle (\hat{a}^\dagger)^2 \rangle = -\sinh r \cosh r$$

Evaluating Mode 1 (squeezed by parameter r):

$$\begin{aligned}
\langle \hat{q}_1^2 \rangle &= \frac{1}{4}\langle \hat{a}_1^2 + (\hat{a}_1^\dagger)^2 + \hat{a}_1\hat{a}_1^\dagger + \hat{a}_1^\dagger\hat{a}_1 \rangle \\
&= \frac{1}{4}\left(-2\sinh r \cosh r + \cosh^2 r + \sinh^2 r\right) \\
&= \frac{1}{4}(\cosh r - \sinh r)^2 = \frac{1}{4}e^{-2r}
\end{aligned}$$

$$\begin{aligned}
\langle \hat{p}_1^2 \rangle &= -\frac{1}{4}\langle \hat{a}_1^2 + (\hat{a}_1^\dagger)^2 - \hat{a}_1\hat{a}_1^\dagger - \hat{a}_1^\dagger\hat{a}_1 \rangle \\
&= -\frac{1}{4}\left(-2\sinh r \cosh r - \cosh^2 r - \sinh^2 r\right) \\
&= \frac{1}{4}(\cosh r + \sinh r)^2 = \frac{1}{4}e^{2r}
\end{aligned}$$

Evaluating Mode 2 (squeezed by parameter $-r$): Substituting $r \to -r$ into the above results directly flips the exponents:

$$\begin{aligned}
\langle \hat{q}_2^2 \rangle &= \frac{1}{4}e^{-2(-r)} = \frac{1}{4}e^{2r} \\
\langle \hat{p}_2^2 \rangle &= \frac{1}{4}e^{2(-r)} = \frac{1}{4}e^{-2r}
\end{aligned}$$

Because the input modes are independent, all cross-correlations between Mode 1 and Mode 2 are zero. The input state $\hat{S}(r)|0\rangle_1 \otimes \hat{S}(-r)|0\rangle_2$ is a product state, making its covariance matrix a block-diagonal tensor sum of the two modes:

$$\sigma_{in} = \frac{1}{4}\begin{pmatrix} e^{-2r} & 0 & 0 & 0 \\ 0 & e^{2r} & 0 & 0 \\ 0 & 0 & e^{2r} & 0 \\ 0 & 0 & 0 & e^{-2r} \end{pmatrix}$$

2.19.2 Deriving the Symplectic Beam-Splitter Matrix

A standard beam-splitter transforms two input modes $(\hat{a}, \hat{b})$ into output modes $(\hat{c}, \hat{d})$. For a balanced 50:50 beam-splitter ($\eta = 1/2$), the canonical transformations are:

$$\hat{c} = \frac{1}{\sqrt{2}}\hat{a} + \frac{1}{\sqrt{2}}\hat{b}$$
$$\hat{d} = -\frac{1}{\sqrt{2}}\hat{a} + \frac{1}{\sqrt{2}}\hat{b}$$

Because the quadratures $\hat{q}$ and $\hat{p}$ are linear combinations of the mode operators, they inherit this exact same transformation geometry. Mapping the inputs $(\hat{q}_1, \hat{p}_1, \hat{q}_2, \hat{p}_2)$ to outputs $(\hat{q}_c, \hat{p}_c, \hat{q}_d, \hat{p}_d)$ produces a system of linear equations:

$$\hat{q}_c = \frac{1}{\sqrt{2}}\hat{q}_1 + \frac{1}{\sqrt{2}}\hat{q}_2$$
$$\hat{p}_c = \frac{1}{\sqrt{2}}\hat{p}_1 + \frac{1}{\sqrt{2}}\hat{p}_2$$
$$\hat{q}_d = -\frac{1}{\sqrt{2}}\hat{q}_1 + \frac{1}{\sqrt{2}}\hat{q}_2$$
$$\hat{p}_d = -\frac{1}{\sqrt{2}}\hat{p}_1 + \frac{1}{\sqrt{2}}\hat{p}_2$$

Expressing this linear transformation as a 4×4 matrix equation $\vec{R}_{out} = B\vec{R}_{in}$ yields the real symplectic matrix B:

$$B = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

2.19.3 Deriving the Output EPR Covariance Matrix

The final covariance matrix evolves under the symplectic transformation as $\sigma_{EPR} = B\sigma_{in}B^T$. We evaluate this matrix product.

Step 3a: Calculate the intermediate matrix $B\sigma_{in}$

$$B\sigma_{in} = \frac{1}{4}\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} e^{-2r} & 0 & 0 & 0 \\ 0 & e^{2r} & 0 & 0 \\ 0 & 0 & e^{2r} & 0 \\ 0 & 0 & 0 & e^{-2r} \end{pmatrix}$$
$$= \frac{1}{4}\begin{pmatrix} \frac{e^{-2r}}{\sqrt{2}} & 0 & \frac{e^{2r}}{\sqrt{2}} & 0 \\ 0 & \frac{e^{2r}}{\sqrt{2}} & 0 & \frac{e^{-2r}}{\sqrt{2}} \\ -\frac{e^{-2r}}{\sqrt{2}} & 0 & \frac{e^{2r}}{\sqrt{2}} & 0 \\ 0 & -\frac{e^{2r}}{\sqrt{2}} & 0 & \frac{e^{-2r}}{\sqrt{2}} \end{pmatrix}$$

Step 3b: Calculate the final matrix $(B\sigma_{in})B^T$

$$(B\sigma_{in})B^T = \frac{1}{4}\begin{pmatrix} \frac{e^{-2r}}{\sqrt{2}} & 0 & \frac{e^{2r}}{\sqrt{2}} & 0 \\ 0 & \frac{e^{2r}}{\sqrt{2}} & 0 & \frac{e^{-2r}}{\sqrt{2}} \\ -\frac{e^{-2r}}{\sqrt{2}} & 0 & \frac{e^{2r}}{\sqrt{2}} & 0 \\ 0 & -\frac{e^{2r}}{\sqrt{2}} & 0 & \frac{e^{-2r}}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

We compute the dot products for the active non-zero elements:

- **Element (1,1) and (3,3):** $\frac{1}{4}\left(\frac{1}{2}e^{-2r} + \frac{1}{2}e^{2r}\right) = \frac{1}{4}\left(\frac{e^{2r}+e^{-2r}}{2}\right) = \frac{1}{4}\cosh 2r$
- **Element (2,2) and (4,4):** $\frac{1}{4}\left(\frac{1}{2}e^{2r} + \frac{1}{2}e^{-2r}\right) = \frac{1}{4}\cosh 2r$
- **Element (1,3) and (3,1):** $\frac{1}{4}\left(-\frac{1}{2}e^{-2r} + \frac{1}{2}e^{2r}\right) = \frac{1}{4}\left(\frac{e^{2r}-e^{-2r}}{2}\right) = \frac{1}{4}\sinh 2r$
- **Element (2,4) and (4,2):** $\frac{1}{4}\left(-\frac{1}{2}e^{2r} + \frac{1}{2}e^{-2r}\right) = \frac{1}{4}\left(\frac{e^{-2r}-e^{2r}}{2}\right) = -\frac{1}{4}\sinh 2r$

Assembling these elements gives the output covariance matrix:

$$\sigma_{EPR} = \frac{1}{4}\begin{pmatrix} \cosh 2r & 0 & \sinh 2r & 0 \\ 0 & \cosh 2r & 0 & -\sinh 2r \\ \sinh 2r & 0 & \cosh 2r & 0 \\ 0 & -\sinh 2r & 0 & \cosh 2r \end{pmatrix} \tag{2.83}$$